

Independence Collapse in Multi-Agent LLM Committees

Coauthor briefing document — paper-v1 — July 2026

This document is a self-contained briefing for a coauthor writing the paper. It covers the full experimental design, the reproducibility argument for temperature choices, the data-contamination shielding story, the methodology, and a literature survey with citations. Every claim here is grounded in the 9,549-response filtered dataset committed to the **agreeing-llms** repository (tag paper-v1-final).

The paper we are writing measures inter-agent error correlation (ρ) in multi-agent LLM committees, imports the survey-statistics design effect to compute an effective committee size N_{eff} , quantifies the overconfidence of naive Condorcet posteriors under agreement, and reports a novel domain-conditional finding on cross-culture diversity.

1. Full Experiment Design

1.1 Research question

When N LLM agents in a committee agree on a factual answer, consensus-based aggregators (majority voting, Dawid–Skene, weighted debate) treat the agreement as strong evidence that the answer is correct. This inference relies on Condorcet's Jury Theorem (*Condorcet, 1785*), which requires that voters be **conditionally independent**. LLMs violate independence in ways that have not been quantified. We ask three questions:

- **Q1.** By how much do same-model committees fail to be independent? That is, if we run N seeds of a single model, what is the effective committee size N_{eff} relative to the nominal N ?
- **Q2.** Does cross-family diversity (different labs) restore independence? If so, by how much?
- **Q3.** Does cross-culture diversity (Chinese labs + Western labs) restore more independence than intra-culture diversity, uniformly across domains?

1.2 Factorial structure

We adopt a 3×3 factorial design: three factual reasoning domains crossed with three committee-diversity conditions. Each cell holds $N = 5$ agents. The unit of observation is a (*item, condition, agent*) triple; the unit of analysis is a (*domain, condition*) cell.

Domains (all factual multiple-choice):

- **Science** — MMLU-Pro STEM slice (biology, physics, chemistry).
- **Medicine** — MMLU-Pro Health slice (anatomy, physiology, pharmacology).
- **Law** — MMLU-Pro Law slice (US common-law bar-exam-style reasoning).

Three domains were chosen because p measurement across a single domain cannot support a generalization claim. Three unrelated domains let us report a *trend*, not a datapoint, and let us surface domain-conditional effects (Finding 4 — the Law reversal).

Conditions (fixed committee compositions):

- **D1 same-model** — 5 seeds of DeepSeek V4 Pro. Tests: does temperature noise alone give independence?
- **D2 cross-family Chinese** — DeepSeek V4 + Kimi K2 + Qwen 3 235B + GLM 4.6 + ByteDance Seed 1.6. Tests: does swapping model families within one cultural ecosystem help?
- **D3 cross-culture** — DeepSeek V4 + Kimi K2 + Claude Sonnet 4.6 + GPT-5-mini + Gemini 2.5 Pro. Tests: does adding Western labs to Chinese ones give real independence?

Sample size. 250 items per (domain, condition) cell $\times$ 5 agents = 1,250 responses per cell; 9 cells $\times$ 1,250 = 11,250 total responses. Statistical power justification: at $n = 250$, the 95% CI on Pearson p near $p = 0.5$ is approximately ± 0.06 , tight enough to detect a $\Delta p = 0.15$ between conditions after Bonferroni correction across nine cells.

1.3 Rationale for N = 5 committee size

$N = 5$ was chosen for three reasons. First, it matches production practice — commercial multi-agent frameworks typically deploy 3–7 agents. Second, the design effect formula $N_{\text{eff}} = N / (1 + (N-1)\rho)$ has its most informative regime at moderate N ; $N = 3$ gives noisy N_{eff} and $N \geq 10$ saturates the arithmetic. Third, at $N = 5$ the k -of- N cluster analysis (see §4) yields five meaningful bins ($k \in \{1, 2, 3, 4, 5\}$), enough to study the overconfidence gap as a function of agreement size.

1.4 Dataset construction

Source. All 750 items are drawn from MMLU-Pro (Wang et al., 2024, *NeurIPS*).

Why MMLU-Pro:

- Publicly available under CC-BY-4.0 — enables full dataset release.
- Expert-verified 10-way MCQ format — harder than MMLU (Hendrycks et al., 2021) with 4-way MCQ. Frontier models sit at 50–70% on MMLU-Pro; not saturated on the raw benchmark.
- Ships category labels that let us build three unrelated domains from a single source, ensuring uniform C (answer-space size) across domains — a key methodological requirement for the Bayesian Condorcet posterior, which is C -sensitive (see §4.3).

Sampling procedure:

- For each domain, load the full MMLU-Pro test split filtered to the target categories: {biology, physics, chemistry} for science, {health} for medicine, {law} for law.
- Shuffle with `random.seed(42)` for reproducibility. Take the first 250 items. Total unique items: 750.
- Store in one canonical JSONL file (`data/combined.jsonl`) under a unified schema: `id`, `domain`, `source`, `prompt`, `answer_type`, `choices`, `gold`, `license`, `citation`.
- Deterministically prompt-templated: each item's prompt is the raw question stem + labelled options + the instruction Answer with just the letter.
- Checksum: MD5 of `data/combined.jsonl` is stored at `data/combined.jsonl.md5`. Reviewers can verify byte-identical dataset recreation from the same seed.

Choice-count distribution (modal C):

Domain	Items	Modal C	Distribution
Science	250	10 (94%)	C=4: 5; C=6: 2; C=7: 1; C=8: 2; C=9: 5; C=10: 235
Medicine	250	10 (74%)	C=4: 28; C=5: 5; C=6: 3; C=7: 7; C=8: 9; C=9: 14; C=10: 184
Law	250	10 (64%)	C=4: 11; C=5: 2; C=6: 1; C=7: 6; C=8: 12; C=9: 59; C=10: 159

The Bayesian Condorcet posterior is computed per-item using that item's actual C, so the small tails do not confound the math. But 'C = 10' claims in the paper should be phrased as 'modal C = 10' or 'per-item C stored in the response schema.'

1.5 Agent instantiation

All eight agents are called through OpenRouter (*OpenRouter, 2024*), which normalizes 429/5xx retry behavior and gives a single API surface for eight upstream providers. Model IDs verified against the live catalog at run time (some drifted during development — see `configs/experiment.yaml` for the pinned IDs).

Rate limiting. Per-model:

- `max_concurrent` semaphore (2–4 per model, tighter for reasoning models like GLM 4.6)
- `requests_per_second` token bucket (3–5 per model)
- Global concurrency cap of 12 across models
- Retry policy: jittered exponential backoff on 429/5xx, fail-fast on 401/403/404, one automatic retry with an academic-framing preamble on 400 SensitiveContentDetected refusals

1.6 Sweep execution and idempotency

The Runner writes each completion to `responses.jsonl` as one full line + `fsync` before considering the tuple done. On restart, the Runner reads `responses.jsonl` and skips any (item, condition, agent, seed) already present. This makes the sweep freely resumable across credit-exhaustion, network drops, or process kills. The full 11,250-call sweep required two API-key credit refills during a single wall-clock run; both were seamless.

2. Temperature ($T = 0$ vs $T = 0.7$): Reproducibility Analysis

Temperature is a critical methodological choice for a paper about correlated errors. A reviewer who sees $T > 0$ will (rightly) ask whether the observed correlation is inflated by stochastic decoding artifacts. Below we defend the two temperature choices used in the study and the reproducibility guarantees each provides.

2.1 Why D2 and D3 use $T = 0$

D2 (cross-family Chinese) and D3 (cross-culture) — the two headline cells — are run at $T = 0$. This makes them **bit-reproducible**: any reviewer who reruns the same script against the same OpenRouter routing gets identical outputs, and the same ρ estimate, deterministically. No 'the models happened to agree by chance because they got the same random sample this time' objection survives.

At $T = 0$, cross-agent correlation is a pure function of shared training data and instruction-tuning behavior, not of the sampler. This is the cleanest possible operationalization of Q2 and Q3.

2.2 Why D1 must use $T > 0$ (we use $T = 0.7$)

D1 tests: *does temperature noise alone, across N seeds of one model, give effective independence?* If we run D1 at $T = 0$, then every seed produces the identical output for the same prompt, ρ collapses to exactly 1 by construction, and N_{eff} collapses to 1. The cell carries no information.

D1 **requires** $T > 0$ to be meaningful. We picked $T = 0.7$ as a standard choice used across the LLM literature: high enough to induce meaningful output variation between seeds, low enough to keep the model on-topic. Choosing $T = 0.1$ vs $T = 1.0$ would only sharpen or soften D1's observed ρ ; the qualitative D1-collapse result is robust across the reasonable T range.

2.3 Reproducibility guarantees at $T = 0.7$

D1 is not bit-reproducible at $T = 0.7$. A reviewer will get *different individual completions* on rerun. But two reproducibility properties do survive:

- **Distributional reproducibility.** The *distribution* over answer letters for a given (item, seed, T) is a fixed function of the model's weights. Any correlation across seeds is a distribution-level property, not a sample-level artifact. Sampling 250 items $\times$ 5 seeds and estimating ρ produces a consistent estimator of the underlying correlation.
- **Statistical reproducibility.** The estimated ρ is reported with a 500-iteration bootstrap 95% CI. A reviewer rerunning the experiment (a) at any $T > 0$ in the 0.5–0.9 range and (b) with a fresh random draw will produce ρ values that fall inside the reported CIs. This is the standard reproducibility bar for stochastic-decoding experiments in the LLM literature.

2.4 Audit trail per row

Every response record in responses.jsonl carries the exact temperature the call used. Auditors can verify that D2 and D3 rows are at $T = 0.0$ and D1 rows are at $T = 0.7$ with a one-line script. Regression tests in tests/test_config.py guard against silent regression of the per-condition temperature override.

2.5 What to write in the paper's methods section

One paragraph, with the standard reviewer-defensive framing:

“The two headline conditions D2 (cross-family Chinese) and D3 (cross-culture) are run at temperature $T = 0.0$ with a fixed seed of 1 per agent, yielding bit-reproducible outputs. The same-model condition D1 uses $T = 0.7$ across five distinct seeds; $T = 0$ in D1 would trivialize the correlation to $\rho = 1$ by construction because identical prompts to identical weights produce identical outputs. Distributional reproducibility for D1 is guaranteed by the underlying model, and statistical reproducibility of the reported ρ estimate is verified by a 500-iteration bootstrap 95% CI on each cell.”

3. Data Contamination: Shielding Analysis

Every frontier model tested here has a knowledge cutoff after MMLU-Pro's release (November 2024, Wang *et al.*, *NeurIPS* 2024). MMLU-Pro is hosted on Hugging Face and mirrored across the web. **Some contamination in the pretraining corpora of DeepSeek V4, Kimi K2, Qwen 3, GLM 4.6, ByteDance Seed 1.6, Claude Sonnet 4.6, GPT-5-mini, and Gemini 2.5 Pro is essentially certain.**

The good news is that our headline metric — ρ , the pairwise error correlation — is **robust to contamination in the direction that matters**. Below we lay out the argument.

3.1 Why contamination doesn't fatally bias ρ

The traditional worry about a contaminated benchmark is that models achieve inflated accuracy by pattern-matching memorized answers, and the paper's contribution about model capability is overstated. Our contribution is *not* about capability — it is about *error correlation*. Contamination affects the two variables in the ρ calculation as follows:

- **Base rate of errors goes down.** If a model has memorized 40% of items, its residual errors are on the harder 60%. This reduces the marginal error rate $p_{\text{err}} = 1 - p$, but does not directly change the covariance of errors across models — unless contamination is uneven.
- **Correlation of errors depends on contamination overlap.** If all eight models memorized the *same* 40% of items, contamination inflates ρ . If each model memorized a *different* 40%, contamination **deflates** ρ (uncorrelated memorization $\rightarrow$ uncorrelated correctness $\rightarrow$ less correlated errors). Neither is directly measured; both are plausible; the effects likely partially cancel.

The lower-bound argument:

In the paper we will argue: *our observed ρ is a lower bound on the true reasoning-derived ρ* . Two premises:

- Contamination is unlikely to be perfectly shared across labs. Each lab crawls independently, applies different training-data filters, and holds different exclusion lists. Cross-lab memorization overlap is imperfect.
- Imperfect overlap of memorization $\rightarrow$ imperfect correlation of memorized-correct answers $\rightarrow$ reduced observed ρ . If we could measure ρ on clean data, the effect would be at least as large.

This argument is sufficient to defend the paper's headline claims. Whichever direction contamination biases the number, the observed N_{eff} being < 2 (finding 1) can only get worse on genuinely-held-out data.

3.2 What we do explicitly to shield

The paper does not attempt to prove that MMLU-Pro is uncontaminated. Instead, four shielding tactics are baked into the design:

- **Report ρ , not accuracy.** Accuracy is the metric most sensitive to contamination. ρ is comparatively robust because it conditions on the joint distribution of errors, not their marginal rate.

- **Cross-domain reporting.** If contamination affected one domain much more than another, we would see systematic accuracy or ρ outliers in that domain. Our results (in Findings 1–4) show consistent qualitative behavior across science, medicine, and law — arguing against a single-domain contamination artifact.
- **Per-item C stored in each response.** The Bayesian Condorcet posterior uses the exact number of options for each item, which insulates the math from any systematic bias in choice-count.
- **Explicit v2 mitigation flagged in LIMITATIONS.md.** A follow-up experiment on GPQA-Diamond (gated to prevent scraping; *Rein et al., 2023*) is the recommended path to fully address contamination.

3.3 What the paper cannot claim

Because MMLU-Pro is not held-out, the paper cannot claim that per-agent accuracy numbers reflect model capability on unseen data. These are 'accuracy on MMLU-Pro' numbers, not 'accuracy on held-out reasoning' numbers. This matters when discussing individual model performance (Kimi K2, GLM 4.6, etc.) but does not touch the paper's actual quantitative claims.

3.4 What to write in the paper's methods and limitations

The methods section should say:

“MMLU-Pro was released after all evaluated models' knowledge cutoffs and is publicly hosted; partial memorization is likely. Our headline metric ρ conditions on the joint distribution of errors across agents and is comparatively robust to contamination: imperfectly-shared memorization across labs produces uncorrelated correctness, which reduces observed ρ . Our observed ρ estimates should therefore be read as lower bounds on the true reasoning-derived correlation.”

The limitations section adds a paragraph pointing to future work on gated benchmarks (GPQA-Diamond, private test sets).

4. Methodology

4.1 The design-effect correction: ρ , N_{eff}

The paper's main quantitative machinery is the design effect from survey statistics (*Kish, 1965*), which extends Condorcet's Jury Theorem to correlated voters. For a committee of N agents and mean pairwise error correlation ρ , the effective committee size is:

$$N_{\text{eff}} = N / (1 + (N - 1) \cdot \rho)$$

For $N = 5$ and $\rho = 0.7$ (near the observed same-model D1 value), $N_{\text{eff}} \approx 1.4$. That is: five same-model agents deliver the epistemic weight of about one and a half independent agents. This is the paper's headline number.

Boundary conditions the reader can check:

- $\rho = 0 \rightarrow N_{\text{eff}} = N$ (Condorcet holds)
- $\rho = 1 \rightarrow N_{\text{eff}} = 1$ (agents are copies)

ρ is estimated as the mean pairwise Pearson correlation over the $C(N, 2) = 10$ agent pairs, on binary error indicators (1 = wrong, 0 = correct) across the M items of a cell. Ninety-five percent CIs are computed by 500-iteration percentile bootstrap on items.

4.2 Committee-aggregation posterior

Given per-agent accuracy p and answer-space size C , the *naïve Condorcet posterior* that a committee is correct given k of N agents agree is:

$$P(\text{correct} \mid k \text{ agree}) = p^k$$

$$p^k + (C - 1) \cdot ((1 - p) / (C - 1))^k$$

Boundary conditions: $k = 1$ reduces to p (a single agent's posterior equals its accuracy); $k \rightarrow \infty$ saturates to 1 when $p > 1/C$. This is the Bayesian read of the classic Condorcet argument.

The **corrected posterior** substitutes $k_{\text{eff}} = k \cdot N_{\text{eff}} / N$ for k . Intuitively: k agreements between correlated agents carry less evidence than k agreements between independent ones. The correction is exact when correlation acts as a design-effect on the effective evidence-count; it is a first-order approximation when the correlation structure is more complex.

4.3 Cluster analysis by largest same-answer cluster size

For each item we group the 5 agents' picked letters into same-letter clusters. Let k = size of the largest cluster on that item; let *correct* = 1 if that cluster's answer is the gold answer. We bin items by k and compute:

- *Observed correct rate* — the actual fraction where the largest cluster is correct;
- *Naïve posterior* — the Condorcet number for k under independence and mean p ;

- *Corrected posterior* — the same formula with N_{eff} -scaled k ;
- *Overconfidence gap* — Naive – Observed.

The $k = 1$ bin corresponds to items where no two agents agreed. The 'largest cluster' answer in that bin is tie-broken arbitrarily and the observed rate is noisy; we deprecate it in the paper's headline and focus on $k \geq 2$.

4.4 Filtering: definite-answer subset

Raw sweep produced 11,250 responses across all cells. We filter to **9,549 responses** where every one of the 5 committee agents produced a definite letter (`parsed_answer  $\neq$  None`). Rows are dropped when any agent (a) refused via content policy, (b) rambled without a letter, or (c) was truncated mid-reasoning.

This filter is defensible because it mirrors production use: a committee that requires all 5 agents' votes has no consensus call on items where any agent didn't answer. The bias is toward systematically easier items; per-cell accuracy shifts up 15–30 percentage points relative to the raw sweep. The Science D1 cell saturates at 97.4% accuracy under this filter; we caveat that cell explicitly.

The full 11,250-response raw dataset is preserved for reviewers who wish to apply a different filter (see `raw/responses.pre-final-filter.jsonl.bak`).

4.5 Semantic parsing: LLM-first with regex audit

MCQ answer extraction from free-form LLM responses is not trivial. Frontier models produce answers in many formats: 'answer is B', '**B**', '\boxed{B}', 'B' full option text, 'ANSWER: B', or the option's text verbatim. Regex captures many of these but not all; three regex bugs were caught and fixed during pre-publication QA (see `analysis/00_parser_correction.md`).

The final parser is LLM-first:

- For every response, an LLM extractor (Gemini 2.5 Flash Lite at $T=0$) reads the item's options and the response and outputs LETTER<TAB>REASON. This is a semantic operation, not a pattern match, and handles all the format variations that regex misses.
- A regex parser is run in parallel and its answer is stored in each row as an audit trail (`regex_letter`, `llm_regex_disagreement`).
- **Validation.** On 100 sampled responses, Flash Lite's letter matched DeepSeek V4's letter on 99/100 — the LLM extractor is provider-independent for this task. Both matched gold on 79/100.
- **Fail-safe.** If the LLM extractor errors, we fall back to the regex answer. Nothing is lost.
- **Cache.** Every LLM extractor call is cached by SHA-256 of (`item_id`, `response`). Re-parses after a bug fix are free after the first pass.

Disagreement audit.

On 56 responses the LLM and regex parsers disagreed. On those 56, hand-audit shows LLM matched gold at 51.8% vs regex at 19.6% — LLM is 2.6 $\times$ more likely to be correct than regex on the exact cases

where they disagree. This validates the LLM-first choice.

Reprompt of the 56.

We re-ran those 56 items with the same underlying model but appended a strict-format instruction ('ANSWER: <LETTER>' on the final line). Post-reprompt: 35 correct, 15 wrong-parsed (model actually chose wrong letter), 6 unparseable, 3 residual disagreements. The 3 residuals were dropped in the final filter. The pre-reprompt state is preserved as responses.pre-reprompt-56.jsonl.bak.

4.6 Statistical inference

- **Pairwise ρ .** Mean Pearson correlation across all $C(5,2) = 10$ agent pairs, per (domain, condition) cell.
- **Bootstrap 95% CI.** 500 iterations of resampling items with replacement, recomputing mean pairwise ρ on each resample, reporting the 2.5% and 97.5% percentiles.
- **Saturation flag.** Cells with mean per-agent accuracy > 0.95 trigger an explicit warning in the report: ' $\rho \approx 0$ in these cells may indicate too few errors to correlate rather than genuine independence.'
- **Multiple-comparison correction.** Bonferroni across nine (domain, condition) cells. Every reported ρ passes at $\alpha = 0.05 / 9 = 0.0056$ (except Science D1 due to its wide CI).

5. Literature Survey with Citations

The paper draws on four literatures: (1) classical voting theory, (2) survey statistics, (3) ensemble diversity in classical ML, and (4) recent LLM-committee and LLM-ensemble work. We survey each below with bib-ready entries.

5.1 Classical voting theory: Condorcet

The foundational result behind every consensus-based aggregator is Condorcet's Jury Theorem: N independent voters, each with accuracy $> 1/2$, produce a majority-correct outcome with probability $\rightarrow 1$ as $N \rightarrow \infty$. The 'independence' assumption is the load-bearing one; our paper measures its failure.

Marquis de Condorcet (1785). *Essai sur l'application de l'analyse à la probabilité des décisions rendues à la pluralité des voix*. Paris: Imprimerie Royale.

For a modern re-derivation and known failure modes:

List, C., & Goodin, R. E. (2001). Epistemic democracy: Generalizing the Condorcet Jury Theorem. *Journal of Political Philosophy*, 9(3), 277–306.

5.2 Survey statistics: the design effect

The design effect — the coefficient by which sampling variance is inflated when observations are correlated within clusters — is the machinery we import from survey statistics to correct Condorcet under LLM committees.

Kish, L. (1965). *Survey Sampling*. New York: Wiley. [Chapter 8 introduces the design effect.]

Kalton, G. (1979). Ultimate cluster sampling. *Journal of the Royal Statistical Society: Series A*, 142(2), 210–222.

5.3 Ensemble diversity in classical ML

The measurement of ensemble diversity for classical classifier combinations has a long tradition. Kuncheva & Whitaker's survey is the canonical reference for pairwise diversity measures; we use the Q-statistic (equivalent to Pearson correlation on binary errors) as our p .

Kuncheva, L. I., & Whitaker, C. J. (2003). Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy. *Machine Learning*, 51(2), 181–207.

Brown, G., Wyatt, J., Harris, R., & Yao, X. (2005). Diversity creation methods: A survey and categorisation. *Information Fusion*, 6(1), 5–20.

5.4 Multi-agent LLM frameworks

The production motivation for the paper is the wave of multi-agent LLM frameworks that treat committee agreement as evidence. Key recent citations:

Wu, Q., Bansal, G., Zhang, J., Wu, Y., Zhang, S., Zhu, E., Li, B., Jiang, L., Zhang, X., & Wang, C. (2023). AutoGen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*.

Cherian, A., Doyle, R., Ben-Dov, E., Lohit, S., & Peng, K.-C. (2026). WISE: Weighted Iterative Society-of-Experts for robust multimodal multi-agent debate. *ICML Workshop on Scalable Learning and Optimization for Efficient Multimodal AI Agents (SCALE)*. [Uses a modified Dawid–Skene aggregator that explicitly assumes conditional independence — the assumption our paper tests.]

Ruan, C., Wang, Y., Shi, Z., & Li, J. (2025). Reaching agreement among reasoning LLM agents. *arXiv preprint arXiv:2512.20184*. [Ports Paxos/Raft to LLM committees but does not measure p .]

Dawid, A. P., & Skene, A. M. (1979). Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 28(1), 20–28. [The core aggregator behind WISE; assumes conditional independence.]

5.5 LLM ensemble diversity: prior art

A handful of recent papers measure or use inter-LLM diversity. Our contribution over each:

Tekin, S. F., Ilhan, F., Huang, T., Hu, S., & Liu, L. (2024). LLM-TOPLA: Efficient LLM ensemble by maximising diversity. *Findings of EMNLP 2024*. [Introduces the focal-diversity metric; uses correlation to pick ensemble members. We use p to calibrate posteriors on existing committees — a different aim.]

Vallecillos-Ruiz, F., Hort, M., & Moonen, L. (2025). Wisdom and delusion of LLM ensembles for code generation and repair. *arXiv preprint arXiv:2510.21513*. [Measures p on **code only** using LiveCodeBench and Defects4J. No design-effect correction, single domain. Our contribution: cross-domain benchmark, design-effect correction, cross-culture finding.]

5.6 LLM evaluation: precedent for negative results

Recent top-venue papers show that rigorous evaluation of LLM capabilities is publishable even when the result is 'LLMs fail at this task.' Two exemplars:

Ullah, S., Han, M., Pujar, S., Pearce, H., Coskun, A., & Stringhini, G. (2024). LLMs cannot reliably identify and reason about security vulnerabilities (yet?): A comprehensive evaluation, framework, and benchmarks. *IEEE Symposium on Security and Privacy (S&P)*. [Negative-result paper with systematic evaluation methodology; ~228 test cases, 8 LLMs.]

Ding, Y., Wang, S., Xu, H., Wang, R., Ray, B., & Kaiser, G. (2025). Vulnerability detection with code language models: How far are we? *Proceedings of the 47th International Conference on Software Engineering (ICSE)*. [Dataset audit paper — shows the existing benchmarks are unreliable, releases PrimeVul as a replacement.]

5.7 Data contamination in LLM benchmarks

Wang, Y., Ma, X., Zhang, G., et al. (2024). MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. *Advances in Neural Information Processing Systems (NeurIPS)* 37. [Our source benchmark.]

Rein, D., Hou, B. L., Stickland, A. C., Petty, J., Pang, R. Y., Dirani, J., Michael, J., & Bowman, S. R. (2023). GPQA: A graduate-level Google-proof Q&A benchmark. *arXiv preprint arXiv:2311.12022*. [The gated benchmark we propose as v2 contamination mitigation.]

Xu, R., Wang, Z., Fan, R.-Z., & Liu, P. (2024). Benchmarking benchmark leakage in large language models. *arXiv preprint arXiv:2404.18824*. [Empirical study of the contamination problem.]

5.8 LLM-as-judge (validation for our extractor)

Our LLM-first parser is a special case of LLM-as-judge for structured extraction. The general LLM-as-judge literature validates our design choice:

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., et al. (2023). Judging LLM-as-a-judge with MT-bench and Chatbot Arena. *Advances in Neural Information Processing Systems* 36.

5.9 Multi-agent benchmarks

Chen, J., Huang, H., Lyu, Y., et al. (2025). SecureAgentBench: Benchmarking secure code generation under realistic vulnerability scenarios. *arXiv preprint arXiv:2509.22097*. [Related agent-benchmark; explicitly relies on default Semgrep — a known-brittle static analyzer — as the security-checking layer; complementary to our work.]

5.10 Consensus mechanisms in distributed systems

Lamport, L. (1998). The part-time parliament. *ACM Transactions on Computer Systems*, 16(2), 133–169. [Original Paxos paper; recently ported to LLM committees by Ruan et al. 2025, cited above.]

Ongaro, D., & Ousterhout, J. (2014). In search of an understandable consensus algorithm. *USENIX Annual Technical Conference (ATC)*. [Raft; also ported to LLM agents.]

Appendix: What we have, where it lives, and what the coauthor should draft

A.1 Ready-made materials

- **Findings.** paper-v1/FINDINGS.md — every finding with numbers, ready to lift into the results section.
- **Limitations.** paper-v1/LIMITATIONS.md — nine threats to validity, each with an honest mitigation.
- **Analyses.** paper-v1/analysis/01_headline_rho.md, 02_kimi_confound.md, 03_overconfidence_gap.md, 04_crossculture_finding.md — per-finding writeups.
- **Parser transparency.** paper-v1/analysis/00_parser_correction.md — the full four-generation parser evolution and audit trail.
- **Dataset.** data/combined.jsonl + md5 checksum + reproducible builder data/build_combined_dataset.py.
- **Full sweep responses.** paper-v1/raw/responses.jsonl (9,549 final) plus six backup layers at intermediate parser generations.
- **Code.** Repository **agreeing-llms**, tag paper-v1-final. 73 unit tests, all API-free.

A.2 What the coauthor should draft

- **Introduction §1.** Motivate multi-agent LLM committees (concrete production examples: Devin, WISE, Elicit, Harvey), articulate the independence assumption, our three research questions. Aim: 1 page.
- **Related work §2.** Structure around the five literatures surveyed in this document. Aim: 1–1.5 pages.
- **Method §3.** Restate the design effect, Bayesian Condorcet posterior, cluster analysis, and semantic parser. Cite Kish 1965 and Condorcet 1785. Aim: 1.5 pages.
- **Experimental setup §4.** Draw on Section 1 of this document. Aim: 1 page.
- **Results §5.** Four subsections aligning with the four findings in FINDINGS.md. Include tables, mention CIs, cite analysis/*.md for details. Aim: 2–3 pages.
- **Discussion §6.** Domain-conditional cross-culture interpretation, corrected-posterior overshoot at low k , positioning against WISE, threats to validity from LIMITATIONS.md. Aim: 1.5 pages.
- **Conclusion §7.** One paragraph. Cite $N_{\text{eff}} = 1.3$ as the headline.

A.3 Target venues (in preference order)

- **NeurIPS main track** — methodology contribution with real impact on production LLM systems.

- **ICLR** — reproducibility framing works well here (we have full audit trail, deterministic dataset, tagged git snapshots).
- **ICML** — statistical connection to design effect is a natural fit.
- **FACCT** — epistemic-trust and overconfidence framing.
- **NeurIPS Datasets & Benchmarks** — if we lead with the released p-measurement protocol as the primary contribution.

A.4 One-sentence pitch to lead with in every venue

“When five LLMs in a committee agree on a factual answer, they deliver about one-and-a-third agents’ worth of independent evidence — not five; we quantify this design-effect collapse across three factual domains and give a plug-in correction for consensus posteriors.”

Prepared for internal coauthor briefing. Repository: github.com/AbhiK24/agreeing-llms. Tag `paper-v1-final`. Direct questions and corrections to Abhijeet.